

# STARTUP ZONE

## Ink From DG Sets



Chakr Innovations team

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**F**ounded by Kushagra Srivastava, Arpit Dhupar and Prateek Sachan, Chakr Innovation has devised a novel technology to capture particulate matter from diesel generators (DGs) without adversely impacting the engine's performance or the environment. Moreover, they do not just dump or burn off the captured carbon. As carbon is a useful resource, they process the captured particulate matter in their device called Chakr Shield and use it to create ink pigment. This pigment is non-toxic and is of the same quality as the ink used in industries.

Chakr Shield provides numerous technological advantages as it:

1. Helps meet the compliance requirements instantly
2. Causes minimal back pressure, which means no adverse impact on performance
3. Requires no additional space; the device can be retrofitted by placing on top of the existing generators

The emission control technology is based on a combination of particle agglomeration through thermophoretic

changes and subsequent capture through a carefully engineered capturing chamber.

The device uses following components:

**Heat exchanger.** Exhaust gas from the diesel generator enters the finned-tube heat exchanger. The heat exchanger is air-cooled and specially designed to cause rapid cooling of the exhaust. It reduces the temperature by 200°C. To ensure that particulate matter does not deposit on the tube walls and reduce the heat transfer coefficient, a spring-based cleaning mechanism is used, which periodically removes the scaled particles.

As the exhaust gas passes through the heat exchanger, it undergoes rapid cooling. Due to rapid cooling, the loaded particles in the exhaust begin to agglomerate. This behaviour is based on thermophoresis (or Ludwig-Soret) effect, which refers to the migration of a colloidal particle or large molecule in a solution in response to a macroscopic temperature gradient.

**Novel capture mechanism.** The structure is designed to ensure maximum capture efficiency while creating negligible back pressure or pressure drop across the capture

mechanism. The pressure drop, called differential pressure ( $\Delta P$ ) between the clean exhaust side and the dirty exhaust side of the device, is one of the most important variables that must be considered in exhaust after treatment design because higher pressure drop results in increased fuel consumption and hence higher energy cost. To ensure negligible back pressure, the area available for air-flow is to be kept constant. This is done by capturing the soot particles by slowly passing the exhaust gases through engineered contours and meshes while interacting with solution.

**Self-cleaning mechanism.** The device has a novel solution-based cleaning mechanism to remove any depositions in real time. Soot particles don't react with the solution but are just suspended in it. This ensures that soot particles don't become airborne again on blowing of wind, handling, etc.

**Collection chamber.** The captured particulate matter is collected in a sealed container, which is connected to the capturing chamber. The collected particulate matter is stored in the chamber till every maintenance cycle and can be replaced after fixed running hours depending on usage. To monitor the collection chamber and assess the need for a maintenance visit, the running hours for diesel generator are monitored through a GSM module; the maintenance requirement is based on the running hours and exhaust properties (particulate loading).

By installing its systems in Delhi NCR, Chakr Innovation has prevented 1500 billion litres of air from getting polluted, and provided clean air for 750,000 people. The team hopes to expand this project further and play its part in building a greener tomorrow.

The team believes that once the technology reaches scale the device